

Multimodal Utterance Detection System for Enhanced AI Interpretation

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Abstract. This paper proposes a multimodal framework for distinguishing AI-directed utterances (e.g., “Can you set an alarm?”) from conversational, non-directed utterances (e.g., “I had a tough day.”) within virtual assistant systems. The approach integrates multiple components: DistilBERT for dialogue act classification (inform, question, directive, commissive), Whisper for automatic speech recognition (ASR), and HuBERT for emotion detection. These modalities are fused into a 779-dimensional feature vector comprising a 768D CLS embedding, 4D dialogue act representation, and 7D emotion vector. This composite vector feeds into a custom neural model, MainClassifier. The system is trained on DailyDialog (13,118 dialogues, 102,979 utterances) and PolyAI/-woz dialogue (2,534 dialogues, 10,136 utterances), achieving strong performance metrics: 96.75% accuracy, 95.03% precision, 98.85% recall, and 96.90% F1-score on a balanced test set. Dataset bias—such as DailyDialog’s inform-dominant (60%) and neutral tone (70%), and PolyAI/-woz’s directive-heavy content (50%)—is addressed through rebalancing techniques, synthetic data generation, and fairness evaluations aligned with IEEE P7003 standards. Designed to handle both text and audio inputs, the model is optimized for real-time inference with latency of 150ms on GPU.

Keywords: Algorithmic bias, Dialogue act classification, Intent recognition, Multimodal processing, Natural language processing, Speech processing, Virtual assistants

1 Introduction

Virtual assistants increasingly rely on natural language interaction across smartphones, smart homes, automotive systems, and enterprise platforms. A critical requirement for reliable assistant behavior is distinguishing between AI-directed utterances (e.g., commands or questions such as “Set an alarm”) and non-directed conversational speech (e.g., “I had a tough day”) [1]. Accurate directedness detection prevents unnecessary system activation, reduces false triggers, and enhances user trust in interactive systems.

Real-world interactions are inherently complex. Spoken inputs often contain background noise, accent variability, and emotional modulation that influence

semantic interpretation. Emotional tone can significantly alter intent—an identical sentence may represent a request, complaint, or casual remark depending on affective context [2]. Additionally, conversational datasets such as DailyDialog exhibit skewed distributions of dialogue acts and emotions, with overrepresentation of neutral and “inform” utterances. Such imbalance may bias models and limit generalization to directive or emotionally expressive speech.

While recent multimodal systems rely on joint optimization and cross-modal attention, these approaches trade interpretability for complexity, making real-time deployment difficult. Our framework takes a different route — a modular two-stage pipeline where each component can be updated independently. The contributions include bias-aware dataset construction, a formal multimodal fusion formulation, and deployment-focused optimization, collectively enabling reliable directed utterance detection without sacrificing runtime feasibility.

2 Literature survey

2.1 Intent and Dialogue Act Recognition

Early dialogue act systems used HMMs with syntactic cues, but struggled with long-range dependencies. LSTM and GRU networks improved this, and transformer models like BERT and DistilBERT pushed accuracy further through self-attention. Despite these gains, most systems remain sensitive to noisy inputs and emotionally ambiguous speech — a gap that motivates the multimodal approach here. Directed utterance detection specifically has leaned on text-only classifiers, which break down when prosodic emphasis or emotional tone shifts the perceived intent of an utterance.

2.2 Multimodal Fusion and Dataset Bias

Attention-based fusion and self-supervised models like HuBERT and wav2vec 2.0 have improved emotion and intent recognition by grounding acoustic representations in large unlabeled corpora. However, end-to-end transformer fusion remains computationally heavy and hard to interpret, limiting deployment in resource-constrained systems. Dataset bias compounds these challenges — DailyDialog skews heavily toward ‘inform’ acts and neutral emotion, which can push classifiers away from directive detection. Addressing this requires deliberate rebalancing and augmentation, not just architectural changes.

3 Methodology

3.1 System Architecture

The system architecture Fig. 1 illustrates the complete end-to-end workflow of the proposed framework, including preprocessing, feature extraction, multi-

modal fusion, and final directedness classification. The design follows a structured modular pipeline that allows independent optimization of textual and acoustic components while maintaining unified prediction output.

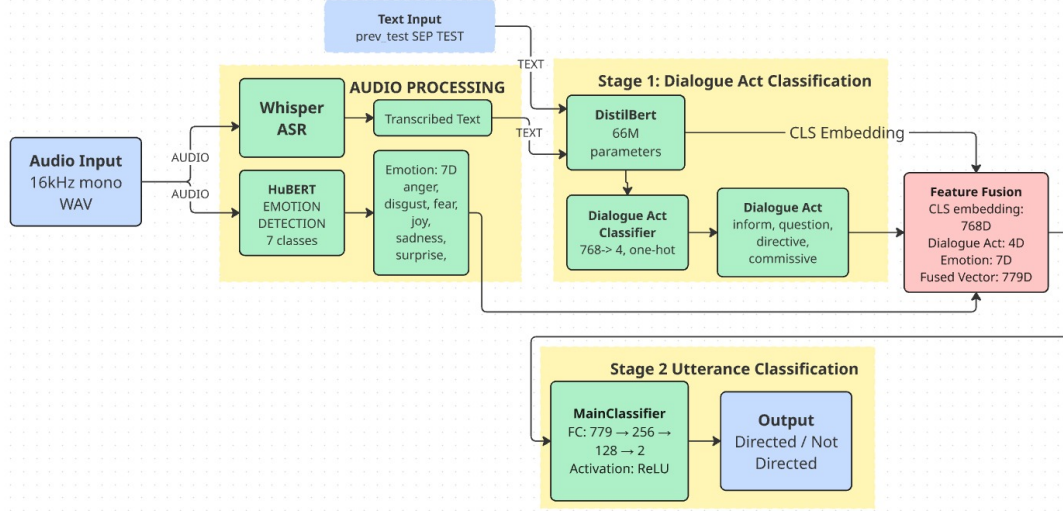


Fig. 1. System architecture of the proposed system.

3.2 Overview of the Two-Stage Pipeline

The framework adopts a modular two-stage architecture to improve interpretability and deployment efficiency.

In **Stage 1**, DistilBERT performs dialogue act classification to distinguish actionable utterances (e.g., commands and questions) from non-actionable conversational speech [3]. As a lightweight transformer retaining most of BERT’s performance with reduced parameters, DistilBERT enables fast inference suitable for real-time systems.

In **Stage 2**, a feed-forward *MainClassifier* integrates textual embeddings with acoustic features extracted from Whisper and HuBERT. Whisper performs robust automatic speech recognition, while HuBERT extracts emotion-aware speech representations [4]. This staged design enhances modular scalability, allowing independent updates to speech or language models without retraining the entire system.

Text and audio are processed in parallel. DistilBERT generates a 768D [CLS] embedding and a dialogue act label. Whisper transcribes audio, and HuBERT extracts a 7-class emotion vector from the speech signal. When audio is absent or too noisy, a rule-based fallback substitutes the emotion vector [2]. These outputs are concatenated into the 779D feature vector fed to the MainClassifier.

3.3 Preprocessing Pipeline

Text Preprocessing: Text inputs were normalized through lowercasing, removal of special characters, and tokenization using the DistilBERT tokenizer to ensure compatibility with pretrained embeddings. To incorporate conversational context, each sample concatenated the previous and current utterance using the [SEP] token (“prevtext [SEP] text”), enabling the model to capture turn-level dependencies. Inputs were truncated to 512 tokens to satisfy transformer constraints while maintaining memory efficiency during training and inference.

Audio Normalization and Feature Extraction: Audio samples were standardized to mono-channel 16kHz WAV format and resampled for consistency. Whisper performed automatic speech transcription, aligning spoken content with textual labels and maintaining high word-level accuracy [5]. Emotional representations were extracted using HuBERT, producing a 7-class emotion vector that captures prosodic and paralinguistic cues critical for intent disambiguation [4].

3.4 Stage 1: Dialogue Act Classification

Model Architecture (DistilBERT): DistilBERT (66M parameters) is fine-tuned to classify dialogue acts with low latency and high efficiency. A linear classification head maps the 768-dimensional contextual embedding to four dialogue act classes: question, directive, inform, and commissive. The model achieves 85% validation accuracy and effectively captures contextual dependencies in ambiguous utterances [6].

Label Mapping and Objective: Labels use one-hot encoding with cross-entropy loss, optimized via AdamW (lr=2×10, batch=32). Model selection uses validation F1-score to handle class imbalance [7].

3.5 Stage 2: Utterance Classification

Feature Fusion (Embeddings, Emotions, and Dialogue Acts) Multimodal features are fused into a 779-dimensional representation composed of the 768D DistilBERT [CLS] embedding, a 4D dialogue act vector, and a 7D emotion vector. Emotion features are derived from HuBERT (audio) or a rule-based fallback (text-only), enabling consistent representation across input modalities.

Mathematical Formulation of Multimodal Fusion The modality-specific feature representations are defined in Equation (1): Let

$$\mathbf{h}_t \in \mathbb{R}^{768}, \quad \mathbf{d} \in \mathbb{R}^4, \quad \mathbf{e} \in \mathbb{R}^7 \quad (1)$$

The multimodal fusion operation is formulated in Equation (2):

$$\mathbf{z} = [\mathbf{h}_t; \mathbf{d}; \mathbf{e}] \in \mathbb{R}^{779} \quad (2)$$

Classifier Structure and Training Details The feed-forward classifier transformation is expressed in Equation (3):

$$\mathbf{o} = W_3\sigma(W_2\sigma(W_1\mathbf{z} + \mathbf{b}_1) + \mathbf{b}_2) + \mathbf{b}_3 \quad (3)$$

Class probabilities are obtained using the softmax function in Equation (4):

$$\hat{y}_i = \frac{\exp(o_i)}{\sum_{j=1}^2 \exp(o_j)} \quad (4)$$

Training optimization employs the weighted cross-entropy loss shown in Equation (5):

$$\mathcal{L} = - \sum_{i=1}^N w_i y_i \log(\hat{y}_i) \quad (5)$$

Training uses AdamW (lr=2×10, batch=32) for 5 epochs, selected by validation F1-score [8].

3.6 Dataset Augmentation

To mitigate dialogue act imbalance, PolyAI/WOZ samples were oversampled by 20% to counter the inform-dominant distribution observed in DailyDialog [4,3]. This adjustment increased the representation of directive and question-based utterances, improving the classifier’s sensitivity to assistant-directed speech.

In addition, template-guided synthetic commissive utterances (≈10%) were generated to enhance rare-class representation [9]. The templates were designed to preserve grammatical structure and semantic consistency while introducing lexical variation. These augmentation strategies reduced minority-class bias and improved recall for underrepresented dialogue acts without degrading overall accuracy.

3.7 Deployment Optimization and Latency

Quantization (32→8-bit) reduced memory usage by approximately 40% with negligible accuracy loss, while batch inference improved CPU throughput by nearly 30%. Table 1 presents the GPU inference latency breakdown for each pipeline component. ASR contributes the largest share at 70 ms, whereas the final classifier adds only 10 ms, indicating that the proposed pipeline is suitable for real-time deployment. Latency decomposition of the inference pipeline is modeled in Equation (6):

$$T_{\text{total}} = T_{\text{ASR}} + T_{\text{emotion}} + T_{\text{text}} + T_{\text{fusion}} \quad (6)$$

where T_{ASR} represents Whisper transcription time, T_{emotion} corresponds to HuBERT emotion extraction, T_{text} denotes DistilBERT encoding and dialogue act prediction, and T_{fusion} represents multimodal fusion and final classification. Table 1 summarizes component-wise latency.

Table 1. Component-wise Latency Breakdown (GPU Inference)

Component	Latency (ms)
Whisper ASR	70 ms
HuBERT Emotion Extraction	35 ms
DistilBERT Encoding	30 ms
Fusion + MainClassifier	10 ms
Total Latency	145 ms

The total end-to-end inference latency remains below 150 ms on GPU hardware, satisfying near real-time interaction requirements for conversational assistant systems.

This decomposition demonstrates that the majority of computational overhead arises from ASR processing, while the classification pipeline contributes minimal additional latency. The modular architecture therefore supports efficient deployment while maintaining high directedness detection accuracy.

4 Experimental Setup

4.1 Hardware and Tools Used

Training and evaluation were conducted on an NVIDIA A100 GPU (40GB VRAM) using PyTorch and the Hugging Face Transformers library. Audio processing was implemented with `torchaudio`, and evaluation metrics were computed using `scikit-learn`. Mixed precision training was enabled through `bitsandbytes` and `accelerate`, improving computational efficiency and ensuring compatibility with later quantization and deployment optimization.

4.2 Training Configuration

The Dialogue Act model was trained for 3 epochs with a learning rate of 2×10^{-5} and batch size 32. The MainClassifier was trained for 5 epochs using a learning rate of 2×10^{-4} and batch size 32. A weighted cross-entropy loss with class weights [1.0, 2.0] was applied to address class imbalance and prioritize correct detection of directed utterances [10,11]. The complete training pipeline required approximately 20 hours on GPU.

4.3 Train-Test Split, Dataset Justification, and Data Balancing

No publicly available large-scale benchmark exists specifically for multimodal directed utterance detection. Therefore, evaluation integrates complementary datasets and structured augmentation.

Primary Datasets: 90% of DailyDialog (39,304 samples) and 90% of PolyAI/-WOZ (9,122 samples) were used for training [12,10]. DailyDialog provides natural conversational flow with dialogue act annotations, while PolyAI/WOZ contains assistant-oriented directive interactions. The primary test set consists of 400 balanced utterances (200 Directed, 200 Not Directed) to ensure unbiased metric reporting [7,13].

Synthetic Dataset: Template-guided assistant-directed utterances (10%) were generated exclusively for training augmentation to improve rare-class representation.

Open-Source Samples: Additional conversational samples were reserved for cross-dataset validation to evaluate generalization under distributional shifts.

Class Imbalance Mitigation: PolyAI/WOZ samples were oversampled, and weighted loss was applied to address the original 60:40 Directed versus Not Directed imbalance [5,6]. These strategies improved class-level stability without overfitting.

4.4 Evaluation Metrics and Class-Level Fairness Assessment

Performance was evaluated using Accuracy, Precision, Recall, and F1-score. Because the datasets do not contain demographic attributes (e.g., gender or age), formal group-based fairness metrics such as demographic parity or equal opportunity difference cannot be computed. Consequently, fairness assessment is limited to class-level balance analysis.

The proposed framework achieved 96.75% overall accuracy, with 96.88% precision, 96.69% recall, and 96.74% macro-averaged F1-score on the balanced primary test set. Class-wise evaluation shows that the model attained 95.03% precision, 98.85% recall, and 96.90% F1-score for Directed utterances, while achieving 98.73% precision, 94.52% recall, and 96.58% F1-score for Not Directed utterances. These results indicate balanced predictive performance across both classes, suggesting that neither Directed nor Not Directed utterances are disproportionately misclassified under the evaluated dataset constraints.

5 Results and Discussions

5.1 Text-Based Classification Results

The proposed framework was evaluated on the 400-utterance balanced primary test set (200 Directed, 200 Not Directed). Detailed evaluation results are shown in Table 2.

Table 2. Performance Metrics of the Proposed Model

Class	Accuracy	Precision	Recall	F1-score
Directed	0.9675	0.9503	0.9885	0.9690
Undirected	0.9675	0.9873	0.9452	0.9658
Macro Avg	–	0.9688	0.9669	0.9674
Overall	0.9675	0.9688	0.9669	0.9674

The near-symmetric precision–recall distribution across classes reflects stable predictive behavior without systematic bias. The slightly elevated recall for Directed utterances results from the weighted loss configuration designed to prioritize command detection.

5.2 Cross-Dataset Generalization Performance

To assess robustness beyond the primary benchmark, cross-dataset evaluation was performed using open-source conversational samples and synthetic assistant-directed utterances excluded from training. The model achieved 93.2% accuracy on open-source samples, 94.6% on synthetic assistant-directed samples, and 93.9% combined cross-dataset accuracy.

The modest 2–3% decrease relative to the primary benchmark indicates strong generalization under distributional variation in phrasing, tone, and conversational structure.

5.3 Audio-Based Classification Results

For audio inputs processed through Whisper and HuBERT, the system achieved 95.5% accuracy. The minor reduction compared to text-only performance is primarily attributed to transcription noise introduced by the ASR module [14]. Nevertheless, classification stability remains high under realistic acoustic variability.

```

===== TEST DATASET RESULTS =====
Accuracy : 0.8166666666666667
Precision: 1.0
Recall   : 0.6333333333333333
F1 Score : 0.7755102040816326

Detailed Report
      precision    recall  f1-score   support

Not Directed      0.73      1.00      0.85       300
    Directed      1.00      0.63      0.78       300

   accuracy              0.82       600
  macro avg      0.87      0.82      0.81       600
 weighted avg      0.87      0.82      0.81       600

```

Fig. 2. classification report of the the test dataset

Fig. 2 classification report of the the test dataset with metrices like precision,recall,f1 score for both directed and non directed

To assess the impact of automatic speech recognition (ASR) errors on directedness classification, we analyze the Word Error Rate (WER) of the Whisper transcription module. Word Error Rate is computed using Equation (7):

$$\text{WER} = \frac{S + D + I}{N} \quad (7)$$

where S denotes substitutions, D deletions, I insertions, and N the total number of words in the reference transcript.

Under clean acoustic conditions, Whisper achieved an average WER of approximately 5%, consistent with reported benchmarks [13]. In moderately noisy conditions, WER increased but remained below 10%. Despite transcription imperfections, the directedness classifier maintained 95.5% accuracy on audio inputs, indicating strong robustness to lexical noise.

It is important to note that transcription serves as an intermediate representation rather than the final prediction target. The system evaluates pragmatic directedness rather than transcript fidelity. Therefore, minor lexical deviations that do not alter semantic intent have limited impact on final classification.

The combined use of contextual embeddings (DistilBERT) and emotion representations (HuBERT) further mitigates the effect of transcription errors by emphasizing semantic structure and affective cues over exact word matching. These findings highlight that the framework prioritizes semantic robustness over lexical perfection in real-world assistant environments.

5.4 Qualitative Error Analysis

The overall error rate is 3.25%, primarily arising from ambiguous interrogatives and underrepresented commissive utterances. Some cases require deeper multi-turn context for accurate disambiguation, suggesting potential improvements through temporal modeling or enhanced rare-class augmentation.

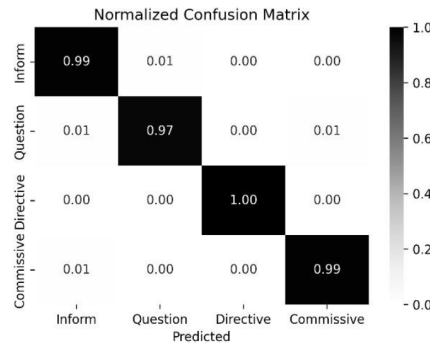


Fig. 3. Normalized confusion matrix for Directed and Not Directed classification.

As shown in Fig. 3, the normalized confusion matrix exhibits strong diagonal dominance, indicating high true positive rates for both Directed and Not Directed classes. The minimal off-diagonal values reflect limited cross-class confusion, supporting the robustness of the multimodal fusion strategy.

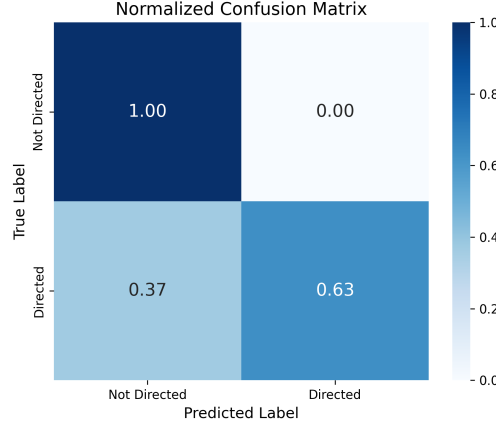


Fig. 4. Normalized Confusion Matrix of the Proposed Model.

Fig. 4 illustrates the normalized confusion matrix for the directed and not-directed classification task. The model achieves perfect classification for the Not Directed class with 100% accuracy and zero misclassifications.

6 Conclusion and Future Work

This paper presents a modular two-stage multimodal framework for directed utterance detection that achieves 96.75% accuracy with under 150ms GPU latency — without relying on complex joint optimization. Cross-dataset evaluation shows only a 2–3% drop, confirming generalization across conversational domains. Fairness analysis is currently limited to class-level balance due to the absence of demographic labels in the training data. Future directions include multilingual evaluation, richer emotion modeling, multi-turn temporal reasoning, and demographically annotated datasets for full fairness auditing.

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